

# Thinh Tran

thinhvt99@gmail.com | 647-515-7345 | github.com/tvtrvn | linkedin.com/in/thinh-tran111

*Full-stack developer shipping production-grade web apps end-to-end — from REST API design and security hardening to data-driven frontends and mobile builds.*

## Education

**Bachelor of Arts in Computer Science (Honours)**, York University, Toronto, ON Sep. 2023 — May 2027

- **Relevant Courses:** Data Structures & Algorithms, Computer Architecture & Organization, Database Management Systems, Software Design Patterns, Object-Oriented Programming, Software Tools, Discrete Mathematics, Applied Linear Algebra, Probability
- **Certifications:** Mastering Design Patterns with Java — CodeSignal (Apr. 2026); Spring Boot REST API with Java & Gradle — CodeSignal (May 2026)

## Work Experience

**Full Stack Developer (On Call)**, Pho Ginger | gingercuisine.ca, Toronto, ON Mar. 2026 — Present

- Built and maintains a production ordering platform (live at **gingercuisine.ca**, ~50 daily customers) with a customer-facing storefront and private staff tablet dashboard; sole developer using **Next.js 16 App Router**, **TypeScript**, **React 19**, **Tailwind CSS v4**, **MongoDB Atlas**, **Prisma 6**, and **Vercel**.
- Developed **13 typed REST API routes** (orders, auth, dashboard CRUD, search, contact, owner menu management, system checks) with Zod schema validation and server-side order-total recomputation to eliminate client-side price tampering.
- Implemented **HMAC-SHA256** signed, HttpOnly, SameSite, Secure session cookies with 12-hour TTL and constant-time comparison; issued per-order 128-bit **viewToken** to prevent confirmation-page enumeration.
- Applied defense-in-depth security: **Upstash Redis** sliding-window rate limits across 5 endpoints, CSRF validation on every state-changing request, and HSTS preload / X-Frame-Options DENY / nosniff / Permissions-Policy headers.
- Executed a **Stripe removal refactor** — deleted ~600 lines of Stripe code (route, webhook, lib, env vars, UI) with zero downtime, migrating order schema to pay-in-person flow.
- Authored a **270-line PRD** covering goals, scope, security model, architecture, and acceptance criteria; documented architecture, API workflows, and deployment steps in full README.

**Inventory Management Associate**, Aritzia Distribution Center, Vaughan, ON Apr. 2025 — Present

- Leveraged warehouse management systems (**WMS**), barcode scanning, and Excel/CSV reporting for precise stock control in a high-volume distribution environment.
- Managed real-time inventory tracking, **SKU accuracy**, and stock reconciliation; implemented cycle counting, variance reporting, and quality checks to maintain data integrity across stored goods.
- Supported operational workflow with strong attention to detail, scaling inventory processes as volume increased.

## Projects

**Portfolio Analytics Dashboard** | GitHub | Live 2026

- Full-stack portfolio analytics platform monitoring 4 simulated portfolios (~\$700M AUM); solo developer using **React 18**, **TypeScript**, **Redux Toolkit** (4 slices), **Recharts** (5 chart types), **Vite 5** (frontend) and **Python 3.11**, **FastAPI**, **SQLAlchemy 2.0**, **Pydantic v2** (backend).
- Designed an **8-table normalized PostgreSQL schema** with 6 composite indexes; built 10 REST API endpoints serving portfolio summaries, holdings, performance series, attribution, risk metrics, and benchmark comparison.
- Implemented a **pure NumPy analytics engine** computing 11 financial metrics from first principles (Sharpe ratio, max drawdown, VaR/CVaR, tracking error, beta, annualized volatility); seeded with geometric Brownian motion across 34 securities in ~850 trading days.

**Historical Trade Scenario Simulator** | GitHub | Live 2025—2026

- Trade simulator for lump-sum and DCA scenarios with benchmark overlays, drawdown chart, per-year return bars, and trade ledger; built with **Vite 5**, **React 18**, **TypeScript**, **Tailwind 3**, **Framer Motion 11**, **FastAPI**, and **PostgreSQL 16**.
- Implemented a **tiered market-data fallback chain** (Tiingo → Yahoo Finance → synthetic GBM) and defense-in-depth middleware: per-IP rate limiting (120 req/min), 64KB body cap, 100 saved-scenarios cap, 30-day TTL cleanup, and a bearer-token-protected maintenance route.
- Achieved **76 passing tests** across 4 suites (finance unit, security end-to-end, maintenance, market-data); automated a **GitHub Actions daily cron** for price refresh and TTL cleanup; enforced security headers (HSTS, X-Frame-Options, COOP/CORP, no-store on all API routes).

**FridgeFit — iOS & Android App** | GitHub 2026

- Cross-platform mobile app built with **Expo SDK 54**, **React Native 0.81**, **React 19**, **TypeScript**, and **Expo Router** (file-based navigation); state managed via **Zustand**, local persistence via **Expo SQLite**, forms via **React Hook Form** + **Zod**.
- Scaffold, design system, navigation, and Fridge tab UI shipped (M1); EAS Build + EAS Submit pipeline configured for App Store / Google Play distribution.

### Top 50 Vietnamese Pop Videos — YouTube Viewer | GitHub

2025

- Next.js 16 App Router app displaying the top 100 most-viewed long-form Vietnamese pop videos on YouTube (25 per page); integrates **YouTube Data API v3** with two parallel search queries, `videos.list` enrichment in batches of 50, and ISO 8601 duration filtering to strip Shorts.
- Implemented a **MongoDB daily-snapshot cache** with a 30-day TTL index to auto-prune stale data; designed quota-aware API usage (~1010 YouTube units per cold load) to stay within the 10,000-unit free daily quota.

### Anti-Procrastination Chrome Extension | GitHub

2025

- Manifest V3 Chrome extension in vanilla HTML/CSS/JavaScript; blocks distracting sites via **declarativeNetRequest** dynamic rules and runs focus sessions with AFK idle detection (`chrome.idle`, 120s threshold).
- Timer uses `chrome.alarms` for service-worker resilience; data persisted via `chrome.storage.local`.

### Connect Four — EECS3311 Software Design | GitHub

2025

- Java MVC game with separated model, view, and controller layers; applied **Observer**, **Strategy**, **Factory**, and **Command** design patterns for maintainable, extensible architecture.
- Wrote **JUnit** unit and integration tests for board rules, invalid moves, save/load, and undo/redo; coordinated development using Git/GitHub, Trello, user stories, and Agile-style sprints.

**ESP32-C3 Embedded LED Firmware:** Programmed firmware for an ESP32-C3 board controlling **WS2812 LEDs** in **C** and **RISC-V Assembly** with **FreeRTOS**; implemented hardware timing control and debugging on a physical development board.

## Technical Skills

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- **Programming Languages:** TypeScript, JavaScript, Python, Java, SQL, HTML/CSS, C/C++, C# (Basic)
- **Frameworks & Libraries:** React (18/19), Next.js 16, React Native, Node.js, FastAPI, Spring Boot, Spring Data JPA, SQLAlchemy 2.0, Prisma 6, Redux Toolkit, Recharts, Tailwind CSS, Framer Motion, React Hook Form, Zod, Pydantic v2, NumPy, Pandas, Pytest, JUnit, JavaFX, Expo, Zustand, Alembic, Upstash Ratelimit, Resend, Gradle
- **Tools & Platforms:** Git, GitHub, GitHub Actions, Docker, Vercel, Koyeb, Netlify, Neon, Expo/EAS, Chrome Manifest V3, Linux/CLI, ESLint, Microsoft Office 365 (Excel, PowerPoint, Word, SharePoint)
- **Databases & Infrastructure:** PostgreSQL, MongoDB, Redis/Upstash, MySQL, Microsoft SQL Server, Azure SQL / Cosmos DB, SQLite; composite indexing, normalized schema design, TTL indexes, serverless cron
- **Concepts & Methodologies:** REST API design, MVC architecture, OOP, Data Structures & Algorithms, Design Patterns (Observer, Strategy, Factory, Command, Singleton, Builder, Adapter, Decorator, Composite), authentication & session management, CSRF mitigation, sliding-window rate limiting, HTTP security headers (HSTS, X-Frame-Options), database indexing & query optimization, serverless cron, mobile-first responsive design, Agile/Scrum, user stories, sprint coordination
- **Testing & Debugging:** Pytest (76 tests, multi-suite), JUnit unit & integration tests, API testing via FastAPI TestClient + `httpx.MockTransport`, ESLint, `tsc -noEmit` type checks, Tectonic CI logging
- **AI & Productivity:** Claude Code, Claude, ChatGPT — agentic workflows, prompt engineering, structured PRD authoring, code review, refactor planning, knowledge-graph-assisted exploration

## Certifications

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- **Mastering Design Patterns with Java** — CodeSignal, Apr. 2026 (verify) — Creational, Structural & Behavioral patterns (Singleton, Factory Method, Builder, Adapter, Decorator, Composite, Observer, Strategy, Command)
- **Spring Boot REST API with Java & Gradle** — CodeSignal, May 2026 (verify) — Spring Core, Dependency Injection, REST Controllers, Spring Data JPA, entity relationships, pagination & sorting

## Additional Information

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- **Languages:** English (Native), French (Fluent), Vietnamese (Advanced)
- **Strengths:** Shipping side projects end-to-end; self-directed learning through hands-on builds; translating non-technical requirements into working software
- **Interests:** Basketball, Volleyball, Rock Climbing, Weightlifting, Cooking, Music